Structure-Preserving Ingestion for Retrieval-Augmented Generation over Research Documents

Alice Chen and Bob Daniels
Kiwi Research Group

Abstract

Retrieval-augmented generation performs poorly on research documents because conventional PDF extraction discards the section hierarchy, tables, and figures that carry a paper's argument. We describe a structure-preserving ingestion pipeline that parses a scholarly PDF into a section tree, extracts figures and tables at their position in the reading flow, and recovers the reference list as structured, identifier-bearing records. We evaluate the pipeline against a fixed-size chunking baseline on a frozen corpus of forty computer science papers and report gains in Recall at five.

1 Introduction

A parsing error propagates through chunking and embedding and surfaces as a confident, wrong answer. This paper argues that parsing, not model choice, is the highest-leverage fix for retrieval over research documents, and describes an ingestion pipeline built around that argument.

The contribution is threefold: a section-aware parser evaluated against academic PDFs, an anchor scheme that survives re-parsing under a different extractor, and an open evaluation harness measuring retrieval quality independently of generation quality.

2 Related Work

Prior work on document parsing for retrieval-augmented generation has focused on layout-aware extraction (Zhao et al., 2025) and vision-guided chunking that segments a page using rendered layout rather than raw text order (Nair, 2026). Structure-fidelity retrieval has separately shown that a single level of section headers is sufficient to keep retrieval localized inside the author-defined structure of a paper (Singh and Okonkwo, 2026).

3 Method

The pipeline has three stages: structural parsing, section-aware chunking, and anchor assignment. Each stage is described below.

3.1 Structural Parsing

A scholarly-document parser converts the source PDF into a section tree, a normalised text stream, and a structured reference list. Figures and tables are extracted with their captions and included at their position in the reading flow rather than collected separately at the end of the document.

3.2 Chunking and Anchoring

Chunk boundaries follow the smallest structural division the parser identifies. Every chunk carries an anchor combining a character offset with a quoted passage, so that a chunk survives a change of parser even though offsets shift.

Strategy	Recall@5
Fixed-size splitting	0.61
Section-aware chunking	0.78

Table 1: Recall at five for each chunking strategy on the frozen evaluation corpus.

4 Experiments

We evaluate retrieval quality on a frozen corpus of forty computer science papers with a golden set of one hundred and twenty query-to-passage pairs. Recall is reported at k of one, three, five, and ten, and Mean Reciprocal Rank is reported as a secondary metric.

5 Results

Section-aware chunking improved Recall at five over fixed-size splitting on the evaluation corpus, with the largest gains observed on papers containing multi-page tables. Table 1 summarises the comparison.

6 Conclusion

Structure-preserving ingestion is a precondition for reliable retrieval over research documents, not an optional refinement. Future work extends the anchor scheme to figures and equations directly rather than their captions alone.

References

Zhao, L., Ma, X., and Chen, R. (2025). Vision-Guided Chunking for Retrieval-Augmented Generation. arXiv:2506.16035.

Nair, P. (2026). SF-RAG: Structure Fidelity for Academic Question Answering. arXiv:2602.13647.

Singh, A. and Okonkwo, T. (2026). Layout-Aware Retrieval for Long Documents. arXiv:2510.07233.

GROBID (2024). A Machine Learning Library for Extracting Bibliographic Information from Scholarly Documents. https://doi.org/10.5281/zenodo.11445660.

Voss, D. and Lin, M. (2025). Empirical Evaluation of PDF Parsing and Chunking for Retrieval-Augmented Generation. arXiv:2604.12047.